

Web Based Diabetes Prediction System Using Machine Learning

Abstract—Diabetes is a common and serious health condition influenced by factors such as age, obesity, physical inactivity, genetic predisposition, unhealthy lifestyle habits, high blood pressure, and poor diet. It raises the risk of severe complications, including heart disease, kidney failure, stroke, vision impairment, and nerve damage. Hospitals currently diagnose diabetes using various medical tests and provide treatment based on the results. Given the massive amount of data available in healthcare, Big Data Analytics plays a key role in detecting hidden patterns and trends that improve diagnostic accuracy and treatment efficiency. Analyzing large datasets helps uncover critical insights that enhance prediction models and lead to better healthcare decisions. This study introduces an advanced predictive model for diabetes classification, integrating additional external factors alongside traditional indicators like glucose levels, BMI, age, and insulin. By incorporating these extra variables, the model significantly improves classification and prediction accuracy over conventional methods. Furthermore, a diabetic pipeline model has been developed to enhance the precision of diabetes detection using advanced data analysis techniques. This data-driven approach enhances diagnostic reliability, improves diabetes classification, and ultimately leads to better treatment strategies. Implementing such models in healthcare enables more effective disease management, earlier detection, and improved patient outcomes, making a significant impact in combating diabetes.

Keywords: Ada Boost, Boosting, Decision Tree, Diabetes, Gradient Descent, IQR, Machine Learning, PIMA dataset, Random Forest, XG Boost

I. INTRODUCTION

Diabetes is a chronic metabolic disorder affecting millions worldwide, characterized by impaired glucose absorption due to insulin deficiency or resistance. According to the International Diabetes Federation, the number of diabetes cases is rapidly increasing due to factors such as urbanization, sedentary lifestyles, and poor dietary habits. In 2016, the WHO reported that 8.5% of adults had diabetes, leading to millions of deaths annually. The condition can cause severe complications, including cardiovascular diseases, kidney failure, and nerve damage, making early detection and management crucial.

Traditional diagnostic methods often face challenges like late detection and misdiagnosis. However, recent advancements in artificial intelligence (AI) and machine learning (ML) have enabled more efficient and accurate disease prediction. AI-powered tools can analyze vast amounts of patient data, helping healthcare professionals detect diabetes at an early stage and personalize treatment plans. This technological shift offers a promising solution to improving diabetes management and reducing its global burden. Diabetes is an endocrine disorder. It exists in various forms, but Type 1 and Type 2 are the most frequent. In Type 1 diabetes, the pancreatic cells are damaged, leading to low insulin secretion. In contrast, Type 2 diabetes involves a combination of low insulin production and insulin resistance. Type 2 is more common and can often be prevented through early detection and lifestyle changes.

II. LITERATURE REVIEW

Aishwarya Mujumdar and Dr. Vaidehi et al.[1] work on various models such as Random Forest, Gradient Boost, SVM, Xg Boost, etc, and AdaBoost in 2019 and finds that AdaBoost gives the best performance

MD. Kamrul Hasan and MD. Ashraful Alam et al. [2] worked on various ML algorithms in 2020 and found that preprocessing and Robust Scaler work well in noisy data, and random forest gives the best result among all.

Muhammad Excell Febrian and Freansiskus Xaverius Fedinan et al.[3] Applied various models on the Pima dataset and concluded that the Naive Bayes Model is good for prediction in 2022

Jingyu Xue, Fancha Min, and Fengying Ma from Jinan, Shandong, China, et al.[4] found that there is no significant relation with age and diabetes, and SVM is the best-performing model for prediction

III. METHODOLOGY

Diabetics can be predicted efficiently and correctly using new technology. The dataset is of 900 rows and 9 columns. We use a total of 6 models: Random Forest, XG Boost, Ada Boost, Gradient Boost, Balanced Random Forest, and Stacking. All the models are trained on 80% of the data and

20% of the data remains for testing purposes. We use Python, Pandas, Numpy, Scikit-learn, matplotlib, pyplot, seaborn and others libraries are used in this model development.

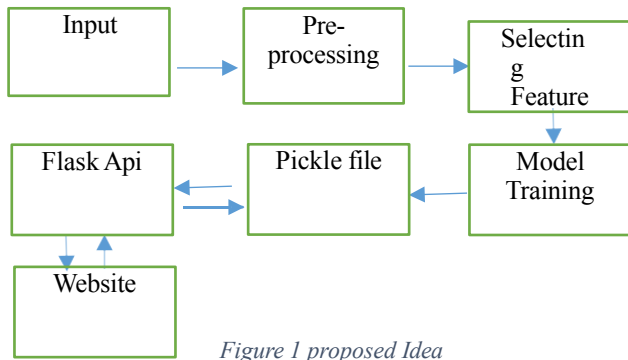


Figure 1 proposed Idea

We evaluate our model through confusion matrix. It is a 2x2 matrix consisting of True Negative(TN), False Positive(FP), False Negative (FN), and True Positive(TP) at 00,01,10,11 positions, respectively. We consider Accuracy, Precision, and Recall to check our model result.

$$\text{Accuracy} = \frac{TP}{TP+FP+FN+TN}$$

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

Based on the result, we choose our best model. After that, we convert the scalar and model into a Pickle file for further integration with the website. Pickle takes a snapshot of the code and Python object and serializes them to a byte stream. Then, when we load it, it deserializes and creates the Python object. Our website is a single-page application, as it was created with React JS. It has 5 Components: Login, SignUp, Home, Diabetic, NonDiabetic. We take all the valuable features from the users and send them to the backend. In the backend, we use Flask, where user authentication and data storage are in MongoDB. Then we load our model then predict the output using the data given by the user. According to the output, we navigate the user to the diabetic or nondiabetic page.

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1

Figure 2: Data Set

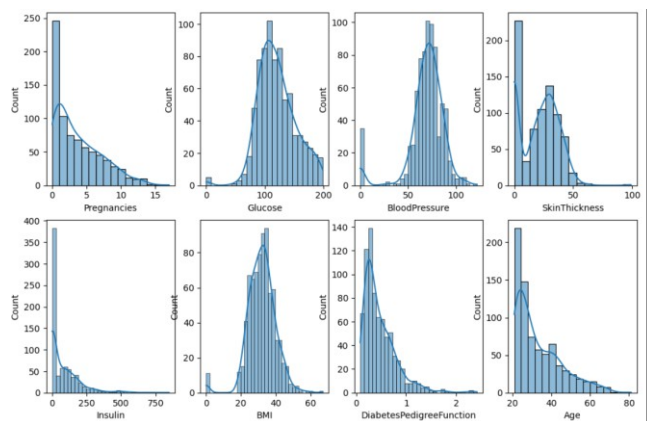


Figure 3 Data distribution before replacing Zero

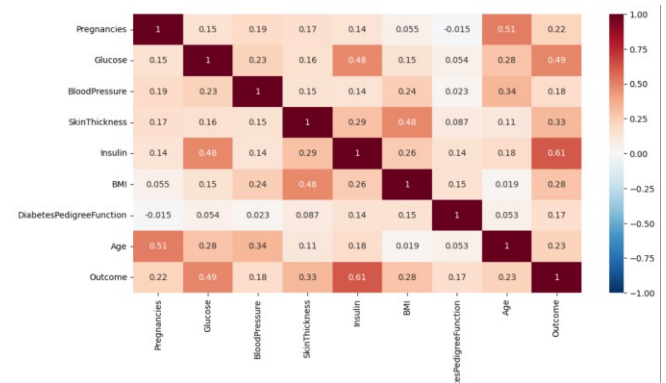


Figure 4: Correlation Matrix

IV. RESULTS

The suggested Methodology will determine whether the patient is prone to diabetes or not. It would help the patient in early detection and take proper precaution.

Algo	Accuracy	Class	Precision	Recall	F1-Score
Random Forest	89%	0	0.93	0.94	0.93
		1	0.87	0.85	0.86
XG Boost	91%	0	0.90	0.96	0.93
		1	0.91	0.80	0.85
Ada Boost	95%	0	0.96	0.96	0.96
		1	0.93	0.93	0.93
Gradient Boost	94%	0	0.96	0.95	0.96
		1	0.90	0.93	0.91
Stacking	92%	0	0.93	0.96	0.94
		1	0.92	0.85	0.88
Balanced Random Forest	91%	0	0.96	0.90	0.93
		1	0.82	0.93	0.87

Figure 5: Accuracy of all models

Among all the models, AdaBoost works efficiently with an accuracy of 96%. AdaBoost works by combining many decision stumps as base learners. In each iteration, it adjusts the weights of the training samples, placing more emphasis on the samples that were misclassified by the previous learner. This iterative process helps the model focus on the harder-to-classify cases. By combining the outputs of all the weak learners through weighted voting, AdaBoost constructs a strong classifier that typically has low bias and controlled variance.

By integrating with a webpage, the user will sign up to the site and then give all necessary input then he will get the result within a fraction of a second. It would boost the productivity.

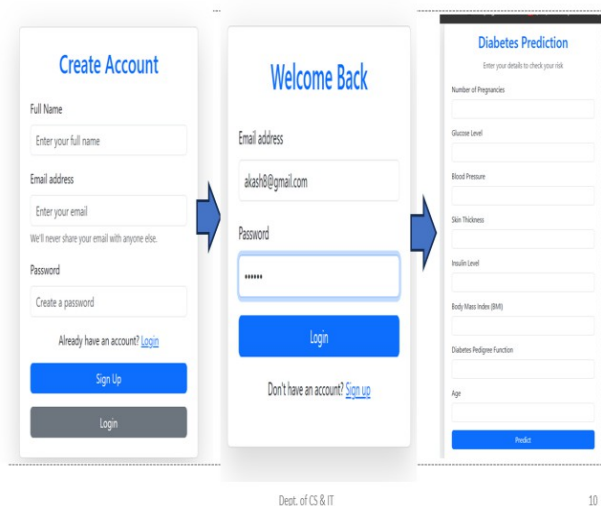


Figure 6 Signup, login, Home Pages

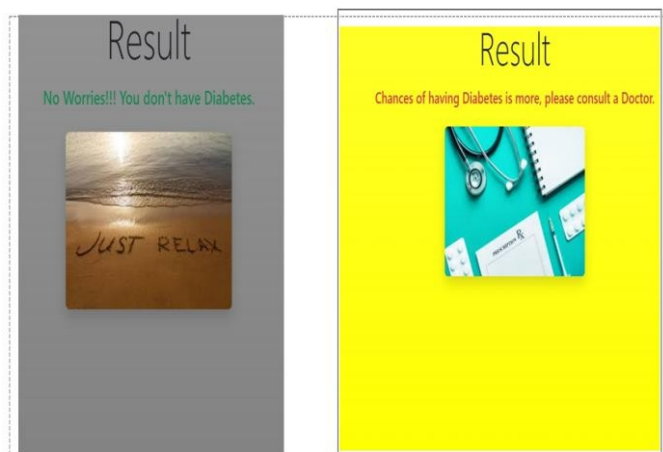


Figure 7: Diabetic Non-Diabetic Pages

V. CONCLUSION

In the health-care business, early detection and accuracy have a great impact on the patient's future. We developed a technology that can accurately predict diabetes, specifically Type 1 and Type 2. In our search, first we clean the data with the IQR method, then add some new features, then use one-hot encoding and raw burst scaler. This data is trained on 6 different models to check different aspects. Among them, AdaBoost does well with some hyperparameter tuning.

We integrate this model into our website, where users can log in and give the required details, and can check the result within a few seconds.

This may help the medical professionals to reduce their workload and support timely. And ease them into treating the patients.

VI. FUTURESCOPE

In the future, we will study a bigger and deeper dataset for patients with more features to increase accuracy. We also want to apply multilayer neural networks to improve accuracy.

we also add a RAG-based chatbot that helps the patients to solve their queries as an assistant medical professional with a trusted source. We can add a reminder for the routine checkup schedule. Provide proper diet and exercise tips using LLMs. Then it can be deployed on Render or AWS for overall use.

VII. REFERENCES

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